

Anesthesia / OT Technician Exam Prep

Part 13 - Solved Question Bank with Detailed Explanations (AIIMS CRE)

1. (Structure of Skeletal Muscles) Which of the following is the basic contractile unit of a skeletal muscle?

- a. Myofibril
- b. Sarcomere
- c. Fascicle
- d. Epimysium

Answer: b) Sarcomere

Explanation: The sarcomere is the basic, fundamental contractile unit of a skeletal muscle. It is defined as the segment between two neighboring Z-lines. Sarcomeres consist of alternating thick (myosin) and thin (actin) protein filaments, which slide past one another to cause muscle contraction according to the sliding filament theory.

2. (Composition of Blood) What percentage of total human blood volume is approximately made up of plasma?

- a. 45%
- b. 55%
- c. 65%
- d. 75%

Answer: b) 55%

Explanation: Human blood is a specialized bodily fluid composed of formed elements (red blood cells, white blood cells, and platelets) suspended in a liquid matrix called plasma. Plasma makes up about 55% of the total blood volume, while the formed elements make up the remaining 45%. Plasma itself is composed of roughly 90-92% water, with proteins, electrolytes, and dissolved gases making up the rest.

3. (Erythrocyte Sedimentation Rate - ESR) An elevated Erythrocyte Sedimentation Rate (ESR) is primarily used as a non-specific clinical indicator for which of the following conditions?

- a. Dehydration
- b. Inflammation
- c. Hypoglycemia
- d. Polycythemia

Answer: b) Inflammation

Explanation: The Erythrocyte Sedimentation Rate (ESR) measures how quickly red blood cells settle at the bottom of a test tube. During inflammatory processes (due to infection, autoimmune disease, or cancer), high proportions of acute-phase proteins like fibrinogen cause RBCs to stick together and settle faster. Thus, a high ESR indicates the presence of systemic inflammation.

4. (Thrombocytopenia) Thrombocytopenia is a medical condition characterized by an abnormally low circulating level of which blood component?

- a. Erythrocytes (Red Blood Cells)
- b. Leukocytes (White Blood Cells)
- c. Platelets (Thrombocytes)
- d. Plasma Proteins

Answer: c) Platelets (Thrombocytes)

Explanation: Thrombocytopenia refers to a deficiency of platelets in the blood. Platelets play a crucial role in blood clotting (hemostasis). A low platelet count can lead to easy bruising, petechiae, and prolonged bleeding from minor cuts. It can be caused by decreased bone marrow production, increased destruction (e.g., ITP), or splenic sequestration.

5. (Functions of Heart) Which chamber of the human heart is responsible for pumping oxygenated blood into the systemic circulation (to the rest of the body)?

- a. Right Atrium
- b. Right Ventricle
- c. Left Atrium
- d. Left Ventricle

Answer: d) Left Ventricle

Explanation: The left ventricle has the thickest muscular wall of all the heart chambers. It receives oxygenated blood from the left atrium and forcefully pumps it through the aortic valve into the aorta, distributing it to the entire systemic circulation. The right ventricle, by contrast, pumps deoxygenated blood to the lungs.

6. (Hypertension) According to typical medical guidelines, persistent blood pressure readings at or above which of the following levels are diagnosed as hypertension in adults?

- a. 110/70 mmHg
- b. 120/80 mmHg
- c. 140/90 mmHg
- d. 100/60 mmHg

Answer: c) 140/90 mmHg

Explanation: Hypertension (high blood pressure) is generally diagnosed when a patient's resting blood pressure is consistently 140/90 mmHg or higher (though recent AHA/ACC guidelines in the US define Stage 1 starting at 130/80 mmHg). Left untreated, hypertension significantly increases the risk of heart failure, myocardial infarction, stroke, and chronic kidney disease.

7. (Physiology of Respiration) During the active process of inspiration (inhalation), what physiological change occurs to the diaphragm?

- a. It relaxes and moves upwards.
- b. It contracts and moves downwards.
- c. It remains stationary while rib muscles contract.
- d. It pushes against the lungs to expel air.

Answer: b) It contracts and moves downwards.

Explanation: Inspiration is an active process. The diaphragm, the primary muscle of respiration, contracts and flattens (moves downwards). This increases the vertical dimension of the thoracic cavity. Concurrently, external intercostal muscles contract, expanding the rib cage. This expansion drops intrathoracic pressure below atmospheric pressure, drawing air into the lungs.

8. (Micturition) The micturition (urination) reflex is primarily mediated and coordinated by which part of the nervous system?

- a. Sympathetic nervous system
- b. Parasympathetic nervous system
- c. Enteric nervous system
- d. Somatic nervous system only

Answer: b) Parasympathetic nervous system

Explanation: The parasympathetic nervous system is primarily responsible for the involuntary aspect of micturition. It innervates the detrusor muscle of the bladder (via pelvic splanchnic nerves), causing it to contract when the bladder is full, while simultaneously relaxing the internal urethral sphincter to allow urine to pass. (Voluntary control is maintained via the somatic nervous system acting on the external sphincter).

9. (KFT Tests) Which of the following biochemical markers is considered the most reliable endogenous indicator of kidney function in a standard Kidney Function Test (KFT)?

- a. Serum Amylase
- b. Serum Creatinine
- c. Blood Glucose
- d. Serum Bilirubin

Answer: b) Serum Creatinine

Explanation: Serum creatinine is a breakdown product of creatine phosphate in muscle. It is freely filtered by the glomerulus and is neither reabsorbed nor actively secreted in large amounts by the renal tubules. Therefore, its level in the blood is inversely related to the Glomerular Filtration Rate (GFR), making it a reliable standard marker for evaluating renal function.

10. (Functions of CSF) Which of the following is NOT considered a primary function of Cerebrospinal Fluid (CSF)?

- a. Providing mechanical buoyancy for the brain
- b. Protecting the brain from sudden impacts
- c. Transporting large amounts of oxygen to neuronal cells
- d. Removing metabolic waste from the central nervous system

Answer: c) Transporting large amounts of oxygen to neuronal cells

Explanation: Transporting oxygen is primarily the function of red blood cells in the cerebral vasculature, not the CSF. The CSF serves to cushion the brain (shock absorption), provide buoyancy (reducing the effective weight of the brain), maintain chemical stability, and clear metabolic waste products from the brain tissue.

11. (Pituitary Gland) Which of the following hormones is synthesized and stored by the posterior pituitary gland?

- a. Growth Hormone (GH)
- b. Thyroid-Stimulating Hormone (TSH)
- c. Antidiuretic Hormone (ADH)
- d. Adrenocorticotrophic Hormone (ACTH)

Answer: c) Antidiuretic Hormone (ADH)

Explanation: The posterior pituitary (neurohypophysis) does not synthesize hormones itself. Instead, it stores and releases hormones that are synthesized in the hypothalamus. These two hormones are Antidiuretic Hormone (ADH or vasopressin) and Oxytocin. GH, TSH, and ACTH are produced by the anterior pituitary gland.

12. (Various Endocrine Hormones) Which endocrine hormone is primarily responsible for regulating the body's basal metabolic rate (BMR)?

- a. Insulin
- b. Thyroxine (T4)
- c. Aldosterone
- d. Parathyroid Hormone (PTH)

Answer: b) Thyroxine (T4)

Explanation: Thyroxine (T4), along with Triiodothyronine (T3), is secreted by the thyroid gland. These thyroid hormones act on nearly every cell in the body to increase cellular metabolism, oxygen consumption, and heat production, thereby primarily regulating the body's Basal Metabolic Rate (BMR).

13. (Routes of Drug Administration) Which route of drug administration typically avoids the first-pass effect and provides 100% immediate bioavailability?

- a. Oral (PO)
- b. Intravenous (IV)
- c. Subcutaneous (SC)
- d. Intramuscular (IM)

Answer: b) Intravenous (IV)

Explanation: The intravenous (IV) route delivers the drug directly into the systemic circulation. By bypassing the gastrointestinal tract and the liver (where the first-pass effect occurs), the entire dose of the drug enters the bloodstream immediately, resulting in 100% bioavailability.

14. (Routes of Drug Elimination) The primary site and route of drug excretion from the human body is through the:

- a. Lungs (exhalation)
- b. Skin (sweat)
- c. Liver (bile)
- d. Kidneys (urine)

Answer: d) Kidneys (urine)

Explanation: While drugs can be eliminated via bile, sweat, saliva, and exhaled air, the kidneys are the primary organs for drug excretion. Water-soluble drugs and drug metabolites (often made more polar by hepatic metabolism) are filtered by the glomeruli or actively secreted into the renal tubules to be eliminated in the urine.

15. (Adverse Drug Reactions) A severe, life-threatening allergic reaction such as anaphylaxis following the administration of a drug is classified as which type of adverse drug reaction?

- a. Type A (Augmented)
- b. Type B (Bizarre/Idiosyncratic)
- c. Type C (Chronic)
- d. Type D (Delayed)

Answer: b) Type B (Bizarre/Idiosyncratic)

Explanation: Type B adverse drug reactions are bizarre, unpredictable, and independent of the drug's dose. They involve hypersensitivity (allergic) reactions, such as anaphylaxis, and idiosyncrasies. In contrast, Type A reactions are dose-dependent augmentations of the drug's known pharmacological effects.

16. (Bronchial Tree) The trachea bifurcates into the right and left primary bronchi at an anatomical landmark known as the:

- a. Larynx
- b. Carina
- c. Glottis
- d. Epiglottis

Answer: b) Carina

Explanation: The trachea extends down into the thorax and bifurcates at a cartilaginous ridge called the carina. This is typically located at the level of the sternal angle and the T4-T5 vertebrae. The carina is highly sensitive, and touching it (e.g., during endotracheal suctioning) can trigger a severe cough reflex.

17. (Bones of Upper Limb) Which of the following bones is NOT structurally a part of the human upper limb?

- a. Humerus
- b. Radius
- c. Tibia
- d. Ulna

Answer: c) Tibia

Explanation: The bones of the upper limb include the humerus (arm), the radius and ulna (forearm), and the carpals, metacarpals, and phalanges (hand/wrist). The tibia (shinbone) is the large bone located in the lower leg, making it part of the lower limb, not the upper limb.

18. (Alkalosis) Respiratory alkalosis is an acid-base disorder primarily caused by:

- a. Hypoventilation (retention of CO₂)
- b. Hyperventilation (excessive loss of CO₂)
- c. Excessive vomiting (loss of stomach acid)
- d. Kidney failure

Answer: b) Hyperventilation (excessive loss of CO₂)

Explanation: Respiratory alkalosis occurs when there is excessive ventilation (hyperventilation) leading to a greater exhalation of carbon dioxide (CO₂) than is being produced by cellular metabolism. This decreases the partial pressure of CO₂ (pCO₂) in the blood, shifting the bicarbonate buffer equation and raising blood pH.

19. (ECG - Electrocardiogram) In a normal sinus rhythm on an Electrocardiogram (ECG), the QRS complex represents which physiological event?

- a. Atrial depolarization
- b. Ventricular depolarization
- c. Ventricular repolarization
- d. Atrial repolarization

Answer: b) Ventricular depolarization

Explanation: The ECG waveform consists of several waves. The P wave represents atrial depolarization. The QRS complex represents the rapid electrical depolarization of the right and left ventricles, which precedes their mechanical contraction. The T wave represents ventricular repolarization (recovery).

20. (Urinary VMA Test) The 24-hour urinary Vanillylmandelic Acid (VMA) test is primarily utilized as a diagnostic marker for which of the following conditions?

- a. Diabetes Mellitus
- b. Pheochromocytoma
- c. Hyperthyroidism
- d. Cushing's Syndrome

Answer: b) Pheochromocytoma

Explanation: Vanillylmandelic acid (VMA) is a major metabolic end-product of catecholamines (epinephrine and norepinephrine). Elevated levels of VMA in a 24-hour urine collection strongly suggest a catecholamine-secreting tumor, most notably a pheochromocytoma, which is a rare tumor of the adrenal medulla that causes severe secondary hypertension.